United States Patent Application Publication	20250271201
Kind Code	A1
Publication Date	August 28, 2025
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DOMESTIC COOLING DEVICE HAVING A PLATE-LIKE SEPARATION UNIT CONFIGURED IN A STORAGE COMPARTMENT

Abstract

A domestic cooling device contains a body, a storage compartment encircled by the body and foods to be cooled can be placed, and a plate-like separation unit arranged in a removable manner in the storage compartment for separation of the volume of the storage compartment and which essentially has a planar base plate. A module carrier assembly is provided which is configured to carry a function module on the plate-like separation unit, particularly to carry a light module. In the domestic cooling device, the module carrier assembly contains a module carrier body which has a longitudinal light module channel arranged such that the light module shall be positioned therein, and an external auxiliary carrier element is embodied in a manner providing passage of at least one light module cable which carries electricity to the light module, and which can be connected to the module carrier body in a removable manner.

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Family ID: **1000008494154**

Appl. No.: **19/056829**

Filed: **February 19, 2025**

Foreign Application Priority Data

TR	2024/002195	Feb. 23, 2024

Publication Classification

Int. Cl.: **F25D23/06** (20060101); **F21V23/00** (20150101); **F21W131/305** (20060101); **F25D27/00** (20060101)

U.S. Cl.:

CPC **F25D23/069** (20130101); **F21V23/002** (20130101); **F25D23/067** (20130101); **F25D27/00** (20130101); F21W2131/305 (20130101); F25D2400/40 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority, under 35 U.S.C. § 119, of Turkish Patent Application TR 2024/002195, filed Feb. 23, 2024; the prior application is herewith incorporated by reference in its entirety.

FIELD AND BACKGROUND OF THE INVENTION

[0002] The present invention relates to a domestic cooling device where a module carrier assembly exists.

[0003] Domestic cooling devices, particularly refrigerators have a storage compartment wherein the foods to be cooled can be placed. There can be at least one plate-like separation unit arranged in a removable manner in the receiving space of the storage compartment. The plate-like separation unit is also arranged for separation of the volume of the storage compartment. The separation unit contains a base plate. Such base plates are also used as a placement plate where objects like food can be placed at the upper side thereof. These base plates are moreover multi-functional units, in this context, these can be used in various manners.

[0004] In the patent document numbered KR 102318549 B1, corresponding to U.S. Pat. Nos. 9,793,763, 10,139,155, 10,312,737, 10,340,741, 10,886,785, 11,239,702, 11,532,954, a refrigerator is disclosed which can illuminate the inner space in an equal manner. This application relates to a refrigerator containing a storage chamber at a predetermined size. In the refrigerator, there is a shelf containing a light source assembled to the storage chamber and embodied to illuminate the storage chamber. In the refrigerator, moreover, there is a transmitter unit connected to an external power supply and embodied to transmit the power in a wireless manner and which has a primary resonance frequency at a predetermined range, and a receiver unit embodied to get power from the transmitter in a wireless manner and to provide this to the light source unit of the shelf.

SUMMARY OF THE INVENTION

[0005] The invention provides an additional improvement, an additional advantage or an alternative to the prior art.

[0006] The object of the invention is to provide a domestic cooling device which has a plate-like separation unit provided in the storage compartment and which has a functional module carrier assembly which is easy-to-produce and which has low-cost.

[0007] The invention, to achieve the abovementioned purpose, is a domestic cooling device containing a body, a storage compartment encircled by the body and wherein the foods to be cooled can be placed, and a plate-like separation unit arranged in a removable manner in the storage compartment for separation of the volume of the storage compartment and which essentially has a planar base plate, and where a module carrier assembly is provided which is configured to carry a function module on the plate-like separation unit, particularly to carry a light module. In the domestic cooling device, the module carrier assembly contains a module carrier body which has a longitudinal light module channel arranged such that the light module shall be positioned therein, and an external auxiliary carrier element embodied in a manner providing passage of at least one light module cable which carries electricity to the light module, and which can be connected to the module carrier body in a removable manner. By means of this, the light module cable can be carried to the light module in a safe manner. Moreover, the light module cable and the electrical socket, connected to the light module cable, are hidden such that the electrical elements like connector are not seen from outside. This provides aesthetically improving the plate-like separation unit which has functional structure. Also, since the auxiliary carrier element can be connected to the module carrier body in a removable manner, an easily reachable structure is provided when necessary (for instance, for maintenance). Moreover, the module carrier body and the removable auxiliary carrier element can be easily adapted to cooling devices which have plate-like separation unit with different widths. In other words, the module carrier body and the removable auxiliary carrier element can be easily scaled in accordance with cooling devices with different widths and can be produced easily and with low cost in a compliant manner.

[0008] Here, the “domestic cooling device” inscription is any device, particularly a refrigerator which is used in houses and which has a cooling cycle and which can realize cooling function.

[0009] Here, the “storage compartment” inscription can mean a volume wherein the foods to be cooled can be placed, particularly a cooling compartment.

[0010] Here, the “base plate” inscription can mean a plate structure which has at least planar surfaces, a glass plate structure made particularly of glass or glass ceramic material.

[0011] Here, the “plate-like separation unit” inscription can mean a separator unit containing a plate structure which at least has the planar surfaces and which functions as a separator wall dividing the storage compartment into more than one volume.

[0012] Here, the “light module cable” inscription can mean an electrical cable which carries electricity to at least the light module. Electrical tools like electrical connector, socket can be connected to the end parts of said electrical cable.

[0013] Here, the “module carrier body” can mean a support body part configured to carry at least the light module.

[0014] Here, the “auxiliary carrier element” inscription can mean a carrier element configured to carry and guide at least one light module cable and which can be connected to the module carrier body in a removable manner.

[0015] In a possible embodiment of the invention, the auxiliary carrier element contains a longitudinal channel which has a cable passage opening through which the light module cable passes and which is opened to the module carrier body from one side. By means of this, the light module cable reaches the module carrier body where the light module is positioned.

[0016] Here, the “longitudinal channel” inscription means a channel provided in the longitudinal direction of the auxiliary carrier element. The longitudinal channel can be defined by means of a base and mutual lateral walls. Again the upper part of the longitudinal channel can be open or can be covered by an upper wall.

[0017] Here, the “cable passage opening” inscription means an opening, a hole or a channel provided in a manner allowing passage of the cable.

[0018] In a possible embodiment of the invention, the module carrier body comprises at least one holding wall embodied such that the auxiliary carrier element can be held at a side which is adjacent to the light module channel, and the auxiliary carrier element has holding channels provided in a manner corresponding to the holding walls. By means of this, the auxiliary carrier element is easily retained to the module carrier body, and the auxiliary carrier element can be easily removed from the module carrier body when required. Moreover, the auxiliary carrier element can be slid in a sliding direction.

[0019] Here, the “holding walls” can essentially be provided in a form which is similar to L-shape, and the holding channels can be configured to be able to held to at least one edge part of said holding walls.

[0020] In a possible embodiment of the invention, the auxiliary carrier element is arranged in a manner extending along a width direction of the holding walls, in order to provide sliding thereof in a sliding direction on the holding walls. By means of this, while the auxiliary carrier element is being slid in the sliding direction on the holding walls, the undesired separation of the auxiliary

carrier element from the holding walls is prevented.
[0021] In a possible embodiment of the invention, the auxiliary carrier element contains a connection extension which extends essentially in a parallel direction with respect to the sliding direction where the auxiliary carrier element is slid on the holding walls, and in a case where the auxiliary carrier element is connected to the module carrier body and extending on the connection extension essentially in an orthogonal direction to the connection extension, there is a connection tab provided in a manner corresponding to a connection housing provided on the module carrier body. By means of this, the auxiliary carrier element is connected to the module carrier body without needing an additional connection element. Also the auxiliary carrier element can be separated from the module carrier body without needing to use an additional tool, for instance, a screwdriver.
[0022] In a possible embodiment of the invention, the auxiliary carrier element contains a channel defining wall embodied in a manner defining the first holding channel, and the channel defining wall has a horizontal connection extension provided in a manner engaging to a longitudinal connection channel embodied on the module carrier body, and in a manner sliding in the connection channel in a condition where the auxiliary carrier element is moved in the sliding direction. By means of this, probable movement irregularities, for instance, yawing is prevented during the movement of the auxiliary carrier element in the sliding direction, and the sliding movement is brought into a more stable and balanced form.
[0023] In a possible embodiment of the invention, the second holding wall is configured to define a cable passage channel for reaching of the light module cable, which passes through the cable passage opening, to the light module channel, and there is an intermediate space in a manner corresponding to the cable passage opening between the first holding wall and the second holding wall for reaching of the light module cable to the cable passage channel through the longitudinal channel. By means of this, the auxiliary carrier element is held to the module carrier body in a balanced manner in the height direction, and at the same time, the light module cable, which passes through the longitudinal channel, reaches to the module carrier body.
[0024] In a possible embodiment of the invention, the auxiliary carrier element contains a bulged wall part which is adjacent to the cable passage opening and provided in a manner guiding the light module cable to the cable passage opening. By means of this, the light module cable is facilitated to be guided to the cable passage opening.
[0025] In a possible embodiment of the invention, the auxiliary carrier element has a casing part provided at a side which is opposite to the side where the connection extension which has the connection tab, is existing and embodied such that an electrical socket to which an end of the light module cable is connected can be placed. By means of this, the auxiliary carrier element is configured to carry also the electrical socket connected to the light module cable.
[0026] In a possible embodiment of the invention, the casing part comprises a socket casing wall provided such that the electrical socket, to which an end of the light module cable is connected, can be rested. By means of this, the electrical socket is prevented from entering into the longitudinal channel.
[0027] In a possible embodiment of the invention, there is at least one stopper extension embodied at a channel input of the longitudinal channel which is close to the casing part and shaped and dimensioned in a manner allowing passage of the light module cable but in a manner preventing passage of the electrical socket. By means of this, passage of the light module cable is allowed, but entry of the electrical socket into the longitudinal channel is additionally prevented.
[0028] In a possible embodiment of the invention, the module carrier body contains a longitudinal module carrier body channel which is adjacent to the light module channel and which extends essentially parallel to the light module channel. By means of this, a reinforced module carrier body is provided which has a light module channel.
[0029] In a possible embodiment of the invention, in the module carrier assembly, there are end

caps which can be retained in a removable manner to the mutual end parts of the module carrier body channel from one side and of the cable passage channel from the other side and shaped and dimensioned in a manner providing at least partially covering the mutual end parts of the module carrier body. By means of this, the end parts of the module carrier body are aesthetically covered.

[0030] In a possible embodiment of the invention, each of the end caps contains a curved extension shaped and dimensioned in accordance with the cable passage channel, and in a condition where the related end cap is retained to the related end part of the cable passage channel, there is a lateral opening, provided for reaching the light module cable to the light module channel, at a side of the curved extension facing the light module channel. By means of this, the end parts of the module carrier body are covered aesthetically in a manner not seen from outside, and at the same time, the end caps prevent reaching of the light module cable to the light module.

[0031] In a possible embodiment of the invention, in order to fix the module carrier assembly to a lower planar surface of the base plate, at least one adhesive element is provided between the lower planar surface and a module carrier planar surface of the module carrier body facing the lower planar surface. By means of this, the module carrier assembly is fixed to the base plate. Here, the adhesive element can be any adhesive element which is solid or liquid-based. Preferably, the adhesive element can be an adhesive band, particularly a double-sided adhesive band.

[0032] In this context, the module carrier body has an extrusion profile structure made by means of an extrusion method. The module carrier body can be made of a plastic material, particularly an acrylonitrile butadiene styrene (ABS) material.

[0033] In this context, the auxiliary carrier element can be made of a plastic material, particularly an acrylonitrile butadiene styrene (ABS) material.

[0034] In this context, the end caps can be made of a plastic material, particularly acrylonitrile butadiene styrene (ABS) or polypropylene (PP) material.

[0035] In this context, the light module casing can be made of a plastic glass material, particularly a non-transparent plexy material, for instance, polymethyl methacrylate (PMMA) material.

[0036] In this context, with the indications “ceiling”, “bottom”, “upper”, “lower”, “front”, “rear”, “horizontal”, “vertical”, “upwards”, “downwards”, “inner”, “outer”, “inwardly”, “outwardly” etc. the positions and orientations given for intended use and intended arrangement of the domestic cooling device and for a user then standing in front of the domestic cooling device in a closed position of the door and viewing in the direction of the domestic cooling device are indicated.

[0037] Each possible embodiment disclosed in this text can be combined with the other possible embodiments disclosed in this text if there is no any technical constraint.

[0038] Other features which are considered as characteristic for the invention are set forth in the appended claims.

[0039] Although the invention is illustrated and described herein as embodied in a domestic cooling device having a plate-like separation unit configured in a storage compartment, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims.

[0040] The construction and method of operation of the invention, however, together with additional objects and advantages thereof will be best understood from the following description of specific embodiments when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0041] FIG. **1** is a diagrammatic, isometric view of a domestic cooling device containing a plate-like separation unit which has a subject matter module carrier assembly;

[0042] FIG. **2** is an isometric view of the plate-like separation unit which has the subject matter module carrier assembly;
[0043] FIG. **3** is an isometric view of the subject matter module carrier assembly, in a condition where the auxiliary carrier element is assembled to the module carrier body;
[0044] FIG. **4** is another isometric view of the subject matter module carrier assembly, in a condition where the auxiliary carrier element is assembled to the module carrier body;
[0045] FIG. **5** is another isometric view of the subject matter module carrier assembly, in a condition where the auxiliary carrier element is not assembled to the module carrier body;
[0046] FIG. **6** is an exploded, isometric view where all parts that exist on the module carrier assembly are shown;
[0047] FIG. **7** is an isometric view of the auxiliary carrier element;
[0048] FIG. **8** is another isometric view of the auxiliary carrier element;
[0049] FIG. **9** is a top view of the auxiliary carrier element;
[0050] FIG. **10** is a top cross-sectional view of the module carrier assembly, in case the end caps are not retained to the module carrier body;
[0051] FIG. **11** is another top cross-sectional view of the module carrier assembly, in case the end caps are not retained to the module carrier body; and
[0052] FIG. **12** is an isometric view of another embodiment of the auxiliary carrier element.

DETAILED DESCRIPTION OF THE INVENTION

[0053] In this detailed explanation, the invented subject matter is explained with examples in order to make the subject more understandable without forming any restrictive effect.
[0054] Within this context, the “width direction x” describes a width direction of a domestic cooling device **1**, a body **2**, a storage compartment **3**, a plate-like separation unit **6**, a base plate **8**, a module carrier assembly **15**, a module carrier body **16**, end caps **18**, **19**, a light module **22** and an auxiliary carrier element **24**.
[0055] Within this context, the “depth direction y” describes a depth direction of the domestic cooling device **1**, the body **2**, the storage compartment **3**, the plate-like separation unit **6**, the base plate **8**, the module carrier assembly **15**, the module carrier body **16**, the end caps **18**, **19**, the light module **22** and the auxiliary carrier element **24**.
[0056] Within this context, the “height direction z” describes a depth direction of the domestic cooling device **1**, the body **2**, the storage compartment **3**, the plate-like separation unit **6**, the base plate **8**, the module carrier assembly **15**, the module carrier body **16**, the end caps **18**, **19**, the light module **22** and the auxiliary carrier element **24**.
[0057] Referring now to the figures of the drawings in detail and first, particularly to FIG. **1** thereof, there is shown a domestic cooling device **1** which has a body **2** which has a lower wall, an upper wall, lateral walls and a rear wall. The body **2** is provided in a manner encircling at least one storage compartment **3** which has a receiving space wherein objects/foods to be cooled can be placed. At a device front side of the domestic cooling device **1**, there is a front opening **4** in a manner providing access to the storage compartment **3** from outside. The front opening **4** is provided at an opposite side of the rear wall of the body **2**. The domestic cooling device **1** moreover has at least one door **5** connected to the body **2** in a manner rotatable by means of hinges. At a closed position where the door **5** is closed, the front opening **4** is completely closed and access to the storage compartment **3** from outside is prevented. As seen in FIG. **1**, in an open position where the door **5** is at least partially open, access to the storage compartment **3** from the front opening **4** can be provided. On the other hand, the body **2** moreover has mutual inner lateral walls **7** which face the storage compartment **3**. In order to separate the volume of the storage compartment **3** as a lower volume and an upper volume, there is a plate-like separation unit **6** arranged in the storage compartment **3** in a removable manner and which essentially has a planar base plate **8** (see FIG. **2**). The plate-like separation unit **6** can be a separator unit which functions as a separator wall inside the storage compartment **3**. The plate-like separation unit **6** is provided in a manner holding

to the inner lateral walls **7**. This holding can be provided by means of bracket structures provided in the plate-like separation unit **6** and by means of similar structures like bracket and wall protrusions provided at inner lateral walls **7**.

[0058] With reference to FIG. **2**, the plate-like separation unit **6** has the base plate **8** which is essentially planar and whereon foods can be placed. The base plate **8** has an upper planar surface **10** whereon the foods are placed, and a lower planar surface **9** at the opposite side of the upper planar surface **10**. There is the first bracket part **11** and the second bracket part **12** extending essentially in parallel direction with respect to each other at sides of the base plate **8** which are opposite to each other. The bracket parts **11**, **12** are provided to extend essentially in the depth direction y. The bracket parts **11**, **12** can be connected to the upper planar surface **10** by means of an adhesive element, for instance, by means of a liquid-based adhesive or a double-sided band not shown in the drawings. The bracket parts **11**, **12** provide holding of the plate-like separation unit **6** to the inner lateral walls **7**. There is a front panel **13** at a front side of the plate-like separation unit **6** which faces the front opening **4**. The front panel **13** has a panel part **14** provided essentially at an orthogonal plane to the base plate **8**. There are front panel connection tools **47** at an inner side of the panel part **14** which faces the base plate **8**. Moreover, as shall be seen in FIG. **2**, there is also a module carrier assembly **15** configured to carry a function module, particularly a light module in the plate-like separation unit **6**. The module carrier assembly **15** is positioned so as to be close to the front panel **13** and so as to be connected to the lower planar surface **9**. The module carrier assembly **15** is configured to extend in the width direction x of the plate-like separation unit **6**, substantially along the whole width of the plate-like separation unit **6**. The module carrier assembly **15** extends at a specific depth in the depth direction y of the plate-like separation unit **6**.

[0059] With reference to FIGS. **3** and **4**, the module carrier assembly **15** essentially contains a module carrier body **16**, an auxiliary carrier element **24**, a light module **22**, a first end cap **18** and a second end cap **19**. The module carrier body **16** can essentially have a profile structure configured such that the light module **22**, the auxiliary carrier element **24**, the end caps **18**, **19** and the front panel **13** are fixed and in a manner carrying these elements. In order to fix the module carrier assembly **15** to the lower planar surface **9** of the base plate **8**, there is at least one adhesive element **20**, **21** provided between the lower planar surface **9** and a module carrier planar surface **17** of the module carrier body **16** facing the lower planar surface **9**. The adhesive elements **20**, **21**, also shown in FIG. **6**, can be the module carrier planar surface **17** from one side and can be a band element, particularly a double-sided band element which can be adhered to the lower planar surface **9** from the other side. On the module carrier planar surface **17**, preferably a first adhesive element **20** and a second adhesive element **21** are provided in a distanced manner to each other and in a manner extending in the width direction x. As shall be seen in FIG. **4**, there are snap-fit claws **36**, **37** at a front side of the module carrier body **16** facing the front panel **13**. In more details, at the front side of the module carrier body **16**, there is a first snap-fit claw **36** and a second snap-fit claw **37** embodied such that the front panel **13** is retained. The snap-fit claws **36**, **37** are shaped and dimensioned such that the corresponding front panel connection tools **47**, provided on the front panel **13**, are at least partially received and held. Preferably, the snap-fit claws **36**, **37** essentially has a form which is similar to C-shape. The front panel connection tools **47** have a form which is essentially similar to cylindrical form in a manner corresponding to the snap-fit claws **36**, **37**. The snap-fit claws **36**, **37** preferably extend in the width direction x of the module carrier body **16**. The front panel connection tools **47** preferably extend in the width direction x of the front panel **13**.

[0060] As shall be seen in FIG. **3**, the module carrier assembly **15** contains an external auxiliary carrier element **24** which can be connected to the module carrier body **16** in a removable manner. The auxiliary carrier element **24** is configured to extend essentially in the width direction x of the module carrier assembly **15**. The auxiliary carrier element **24** is provided in a manner holding to a rear side provided at the opposite side of the front side of the module carrier assembly **15** to which the front panel **13** is connected. In other words, the front panel **13** and the auxiliary carrier element

24 are positioned at opposite sides of the module carrier body **16**. The module carrier body **16** has at least one holding wall **28**, **29** embodied such that the auxiliary carrier element **24** can hold. In more details, the module carrier body **16** has a first holding wall **28** and a second holding wall **29** extending in the width direction x. The holding walls **28**, **29** are positioned in a distanced manner with respect to each other in a height direction z of the module carrier body **16**. The holding walls **28**, **29** are embodied such that there is an intermediate space **63** in between. The holding walls **28**, **29** can have a form which is essentially similar to L-shape.

[0061] As shall be seen in FIGS. **4** and **6**, the light module **22** is positioned in a light module channel **34** embodied in the module carrier body **16** and which essentially extends longitudinally. In other words, the module carrier body **16** has a longitudinal light module channel **34** arranged such that the light module **22** is positioned therein. At a lower side of the module carrier body **16**, there is a longitudinal lower opening **35** that is opened to the light module channel **34**. The light module **22** has a light module casing **54** that exists at the outermost part. Inside the light module casing **54**, as shall be seen in the cross-sectional view as in FIG. **10**, there is the circuit board which has a light source **53** that extends longitudinally. There can be pluralities of light source elements, particularly light emitting diodes LED through the circuit board which has light source **53**. The electrical connection with the circuit board which has the light source **53** provided in the light module casing **54** is provided from a socket input **23** provided on the light module casing **54**. The socket input **23** is embodied at an end part of the light module casing **54**. The light module casing **54** is shaped and dimensioned in a manner placing to the light module channel **34**. There is at least one light module casing connection housing **59** on the light module casing **54**. The light module casing connection housing **59** extends longitudinally on the light module casing **54**. Preferably, at least one couple of light module casing connection housing **59** is embodied at the opposite sides of the light module casing **54**. In the light module channel **34**, there are longitudinal module carrier body connection beams **60** embodied in a manner corresponding to the light module casing connection housing **59**. When the light module **22** is slid and driven inside the light module channel **34**, the module carrier body connection beams **60** are placed to the light module casing connection housings **59**, and by means of this, the light module **22** is retained in the light module channel **34**. In the invention, the light module channel **34** that exists on the module carrier body **16** is arranged at a region between the side where the holding walls **28**, **29** exist, and a module carrier body channel **49**. The module carrier body channel **49** is embodied as a longitudinal channel which is adjacent to the light module channel **34** of the module carrier body **16** and which essentially extends in a parallel manner to the light module channel **34**.

[0062] As shall be seen in FIG. **3** and FIG. **5**, the auxiliary carrier element **24** is configured to provide sliding thereof in a sliding direction s**1**, s**2** in the width direction x. In order to provide sliding of the auxiliary carrier element **24** on the holding walls **28**, **29**, the auxiliary carrier element **24** is arranged to extend along the width direction x of the holding walls **28**, **29**. When the auxiliary carrier element **24** is slid and moved at a first sliding direction s**1**, it moves on the holding walls **28**, **29** and is finally connected to the module carrier body **16**. When the auxiliary carrier element is moved in a second sliding direction s**2**, it again moves by sliding on the holding walls **28**, **29**, and finally, it completely separates from the module carrier body **16**.

[0063] With reference to FIGS. **7** and **8**, the auxiliary carrier element **24** contains a longitudinal channel **25** through which at least one light module cable c passes. The longitudinal channel **24** is defined by the mutual lateral walls **38** of the auxiliary carrier element **24**. A base part of the longitudinal channel **25** can be inclined in a manner guiding the light module cable c. The longitudinal channel **25** has a cable passage opening **30** opened to the module carrier body **16** from one side. The longitudinal channel **25** can preferably have a form which is essentially similar to L-shape. The auxiliary carrier element **24** contains a connection extension **31** which essentially extends in a parallel direction in accordance with the sliding direction s**1**, s**2** where the auxiliary carrier element **24** is slid on the holding walls **28**, **29**. As shall be seen in FIGS. **11** and **12**, the connection tab **32** is provided in a manner corresponding to a connection housing **33** provided on the module carrier body **16** in a condition where the auxiliary carrier element **24** is connected to the module carrier body **16**. By means of this, the auxiliary carrier element **24** is fixed to the module carrier body **16** at the desired position, and after the auxiliary carrier element **24** is assembled to the module carrier body **16**, the undesired probable displacements are prevented. On the other hand, when the auxiliary carrier element **24** is connected to the module carrier body **16**, the cable passage opening **30** corresponds to the intermediate space **63**. By means of this, as shall be seen in FIG. **11**, passage of the light module cable c, which passes through the longitudinal channel **25**, to the cable passage channel **48** that exists in the module carrier body **16** is provided.

[0064] As described before and as shall be seen in FIG. **3**, the module carrier body **16** has holding walls **28**, **29** embodied such that the auxiliary carrier element **24** can be held at a side which is adjacent to the light module channel **34**. Again with reference to FIGS. **7** and **8**, the auxiliary carrier element **24** has holding channels **39**, **41** provided in a manner corresponding to the holding walls **28**, **29**. The auxiliary carrier element **24** comprises a channel defining wall **45** embodied to define the first holding channel **39**. The first holding channel **39** is configured to hold to the first holding wall **28** which exists in the module carrier body. In other words, at least one part of the first holding wall **28** engages to the first holding channel **39**. The channel defining wall **45** also has the horizontal connection extension **42** provided in a manner engaging to a longitudinal connection channel **55** embodied on the module carrier body **16** and in a manner sliding in the connection channel **55** in a case where the auxiliary carrier element **24** is moved in the sliding direction s**1**, s**2**. A condition where the horizontal connection extension **42** engages to the connection channel **55** is seen in FIG. **3**. The connection channel **55** is defined by a body horizontal extension **56** which essentially extends in the width direction x. The channel defining wall **45** and the horizontal connection extension **42** are embodied to extend essentially in the width direction x of the auxiliary carrier element **24** and the module carrier body **16**. The horizontal connection extension **42** also increases the rigidity of the lateral wall of the auxiliary carrier element **24**. In other words, the auxiliary carrier element **24** has a support function for the lateral wall. As shall be seen in FIG. **7** and FIG. **8**, there is also at least one second holding channel **41** on the auxiliary carrier element **24**. Preferably, there can be more than one second holding channel **41** on the auxiliary carrier element **24**. The second holding channels **41** are defined by the holding tools **40**. Preferably, more than one holding tool **40** is embodied to be positioned in a distanced manner with respect to each other in the width direction x such that each holding tool **40** defines the second holding channel **41**. The second holding channels **41** are configured to hold to the second holding wall **29** that exists in the module carrier body **16**. In other words, at least one corresponding related part of the second holding wall **29** engages into the second holding channels **40**. Here, the second holding wall **29** is configured to define a cable passage channel **48** for reaching of the light module cable c which passes through the cable passage opening **30**, to the light module channel **34**. As shall be seen in FIGS. **10** and **11**, the light module cable c, which passes through the cable passage opening **30** and through the intermediate space **63**, reaches the cable passage channel **48** defined by the second holding wall **29**. On the other hand, as shall be seen in FIGS. **8** and **9**, the auxiliary carrier element **24** also comprises a bulged wall part **43** which is adjacent to the cable passage opening **30** and which is in bulged structure and provided in a manner guiding the light module cable c to the cable passage opening **30**. The bulged wall part **43** is embodied to realize a bulge towards the longitudinal channel **25**. By means of this, the bulged wall part **43** has been formed in a manner narrowing the part of the longitudinal channel **25** which is close to the cable passage opening **30**. This facilitates the movement of the light module cable c inside the longitudinal channel **25** after the auxiliary carrier element **24** is assembled to the light module body **16**.

[0065] As shall be seen in FIGS. **7** and **8**, the auxiliary carrier element **24** has a casing part **26** embodied such that an electrical socket not shown in the drawings can be placed to which an end of the light module cable c is connected. The casing part **26** exists at a side which is opposite to the side where the connection extension **31** which has the connection tab **32** is existing. In other words, the casing part **26** is embodied at a side where a channel input **61** is existing where the light module cable c enters the longitudinal channel **25** of the auxiliary carrier element **24**. As shall be seen in FIG. **11**, electrical socket is connected to a first end part c**1** of the light module cable c. The electrical socket is positioned at the casing part **26**. A casing space **46** is defined such that the electrical socket shall be positioned inside the casing part **26**. The casing part **26** is essentially embodied in a box form such that a front part and an upper part are open. There is a casing opening **27** at the front part of the casing part **26**. As shall be seen in FIG. **8**, a base part where the electrical socket is seating in the casing part **26** is embodied so as to be lower in a height direction z with respect to the longitudinal channel **25**. In other words, there is a height difference between the casing part **26** and the longitudinal channel **25**. Thanks to this height difference, a socket casing wall **44** is defined. The socket casing wall **44** functions as a stopper for the electrical socket, and is embodied to prevent entering of the electrical socket into the longitudinal channel **25** in an undesired manner. This aim is obtained by resting the electrical socket to the socket casing wall **44**.

[0066] With reference to FIGS. **4-6**, the end caps **18**, **19** are retained to a module carrier body channel input **50** of the module carrier body **16** in a manner providing at least partially covering of the mutual lateral parts of the module carrier body **16**. In more details, in the module carrier assembly **15**, there are end caps **18**, **19** which can be retained in a removable manner to the mutual end parts of the module carrier body channel **49** from one side and of the cable passage channel **48** from the other side and shaped and dimensioned in a manner providing at least partially covering the mutual end parts of the module carrier body **16**. In other words, there is a first end cap **18** provided in a manner at least partially covering a first end part of the module carrier body **16**, and a second end cap **19** provided in a manner at least partially covering a second end part of the module carrier body **16**. At each of the end caps **18**, **19**, first end cap retaining tools **51** are embodied on the surfaces of the module carrier body **16** facing the related end parts. The first end cap retaining tools **51** are shaped and dimensions in a compliant manner to the module carrier body channel **49**. The first end cap retaining tools **51** are arranged in a manner protruding from the surfaces thereof which faces the related end parts of the module carrier body **16**. The first end cap retaining tools **51** are retained to the related module carrier body channel input **50** of the module carrier body channel **49**, see FIG. **10**. On the other hand, in each of the end caps **18**, **19**, on the surfaces facing the related end parts, there is also the second end cap retaining tools **52**. The second end cap retaining tools **52** are retained to the related end part of a cable passage channel **48** defined by the second holding wall **29**, see FIG. **10**. Each of the end caps **18**, **19** also comprises a curved extension **57** shaped and dimensioned in accordance with the second holding wall **29** and the cable passage channel **48**. The curved extension **57** essentially has a form which is similar to C-shape. In a condition where the related end cap **18**, **19** is retained to the related end part of the cable passage channel **48**, at a side of the curved extension **57** facing the light module channel **34**, there is a lateral opening **58** provided for reaching the light module cable c to the light module channel **34**. Thanks to this lateral opening **58**, the light module cable c which passes through the cable passage channel **48** exits the cable passage channel **48** thanks to the lateral opening **58**, and the light module cable c reaches the light module channel **34**. There is an electrical connector not shown in the drawings at a second end part c**2** of the light module cable c which reaches the light module channel **34**. The electrical connector is connected to the socket input **23** of the light module **22**. By means of this, the electrical energy required for operation of the light module **22** is carried to the light module **22** by means of the light module cable c.

[0067] For exemplification, in FIG. **12**, an isometric view related to another embodiment of the auxiliary carrier element **24** is given. In FIGS. **3-11**, there are drawings where the first embodiment of the auxiliary carrier element **24** exists in the module carrier assembly **15**. All explanations and descriptions, related to the abovementioned auxiliary carrier element **24**, the first embodiment thereof and the module carrier assembly **15**, are also valid for the second embodiment of the auxiliary carrier assembly **24** given in FIG. **12**. The second embodiment of the auxiliary carrier element **24** basically has two differences from the first embodiment of the auxiliary carrier element **24**. The first one of these is provided such that the socket casing wall **44** of the casing part **26** is inclined. By means of this, guidance of the light module cable c to the longitudinal channel **25** is facilitated. The second one of these is that there is at least one stopper extension **62** at the channel input **61** of the longitudinal channel **25**. At the channel input **61**, there are preferably couple of stopper extensions **62** positioned mutually to each other. The stopper extensions **62** are shaped and dimensioned in a manner allowing passage of the light module cable c but in a manner preventing passage of the electrical socket. The stopper extensions **62** can essentially have a form which is similar to L-shape. As the stopper extensions **62** are positioned at the channel input **61**, a structure which is similar to M-shape is formed.

[0068] As seen in FIG. **11**, the path followed by the light module cable c in the light module assembly **15** is mainly as follows: the first end part c**1** of the light module cable c where the electrical socket exists is existing at the casing part **26**. The light module cable c passes through the longitudinal channel that exists in the auxiliary carrier element **24**, and is guided to the cable passage opening **30** thanks to the bulged wall part **43**. Thanks to the intermediate space **63** which corresponds to the cable passage opening **30**, the light module cable c which reaches the cable passage channel **48** passes through the cable passage channel **48** in a manner reaching the socket input **23** of the light module **22**, namely it approaches towards the first end cap **18**. Thanks to the lateral opening **58** that exists on the first end cap **18**, the light module cable c which exits the cable passage channel **48** realizes a cycle and reaches the light module channel **34**. Finally, the second end part c**2** of the light module cable c where the electrical connector is provided is connected to the socket input **23** of the light module **22** that exists in the light module channel **34**. By means of this, the route followed by the light module cable c between the casing part **26** and the socket input **23** is completed.

[0069] The following is a summary list of reference numerals and the corresponding structure used in the above description of the invention: [0070] **1** domestic cooling device [0071] **2** body [0072] **3** storage compartment [0073] **4** front opening [0074] **5** door [0075] **6** plate-like separation unit [0076] **7** inner lateral wall [0077] **8** base plate [0078] **9** lower planar surface [0079] **10** upper planar surface [0080] **11** first bracket part [0081] **12** second bracket part [0082] **13** front panel [0083] **14** panel part [0084] **15** module carrier assembly [0085] **16** module carrier body [0086] **17** module carrier planar surface [0087] **18** first end cap [0088] **19** second end cap [0089] **20** first adhesive element [0090] **21** second adhesive element [0091] **22** light module [0092] **23** socket input [0093] **24** auxiliary carrier element [0094] **25** longitudinal channel [0095] **26** casing part [0096] **27** casing opening [0097] **28** first holding wall [0098] **29** second holding wall [0099] **30** cable passage opening [0100] **31** connection extension [0101] **32** connection tab [0102] **33** connection housing [0103] **34** light module channel [0104] **35** longitudinal lower opening [0105] **36** first snap-fit claw [0106] **37** second snap-fit claw [0107] **38** lateral wall [0108] **39** first holding channel [0109] **40** holding tool [0110] **41** second holding channel [0111] **42** horizontal connection extension [0112] **43** bulged wall part [0113] **44** socket casing wall [0114] **45** channel defining wall [0115] **46** casing space [0116] **47** front panel connection tools [0117] **48** cable passage channel [0118] **49** module carrier body channel [0119] **50** module carrier body channel input [0120] **51** first end cap retaining tool [0121] **52** second end cap retaining tool [0122] **53** circuit board which has a light source [0123] **54** light module casing [0124] **55** connection channel [0125] **56** body lateral extension [0126] **57** curved extension [0127] **58** lateral opening [0128] **59** light module casing connection housing [0129] **60** module carrier body connection beam [0130] **61** channel input [0131] **62** stopper extension [0132] **63** intermediate space [0133] s**1** first sliding direction [0134] s**2** second sliding direction [0135] c light module cable [0136] c**1** first end part [0137] c**2** second end part [0138] x width direction [0139] y depth direction [0140] z height direction

Claims

1. A domestic cooling device, comprising: a body; a storage compartment encircled by said body and where foods to be cooled are placed; at least one light module cable; and a plate-shaped separation unit disposed in a removable manner in said storage compartment for separation of a volume of said storage compartment, said plate-shaped separation unit having a planar base plate and a module carrier assembly configured to carry a function module being a light module on said plate-shaped separation unit, said module carrier assembly having a module carrier body with a longitudinal light module channel disposed such that said light module is positioned therein, and an external auxiliary carrier element embodied in a manner providing passage of said at least one light module cable carrying electricity to said light module, and said external auxiliary carrier element being connected to said module carrier body in a removable manner.

2. The domestic cooling device according to claim 1, wherein said auxiliary carrier element has a longitudinal channel which has a cable passage opening formed therein and through which said at least one light module cable passes and which is opened to said module carrier body from one side.

3. The domestic cooling device according to claim 2, wherein said module carrier body has holding walls and at least one of said holding walls is embodied such that said auxiliary carrier element is held at a side which is adjacent to said longitudinal light module channel, and said external auxiliary carrier element has holding channels provided in a manner corresponding to said holding walls.

4. The domestic cooling device according to claim 3, wherein said external auxiliary carrier element is disposed in a manner extending along a width direction of said holding walls, in order to provide sliding thereof in a sliding direction on said holding walls.

5. The domestic cooling device according to claim 4, wherein: said module carrier body having a connection housing; and said external auxiliary carrier element has a connection extension which extends in a parallel direction with respect to the sliding direction where said external auxiliary carrier element is slid on said holding walls, and in a case where said external auxiliary carrier element is connected to said module carrier body and extending on said connection extension generally in an orthogonal direction to said connection extension, there is a connection tab provided in a manner corresponding to said connection housing of said module carrier body.

6. The domestic cooling device according to claim 4, wherein: said module carrier body having a longitudinal connection channel formed therein; and said external auxiliary carrier element has a channel defining wall embodied in a manner defining a first of said holding channels, and said channel defining wall has a horizontal connection extension provided in a manner engaging to said longitudinal connection channel embodied on said module carrier body, and in a manner sliding in said longitudinal connection channel in a condition where said external auxiliary carrier element is moved in the sliding direction.

7. The domestic cooling device according to claim 3, wherein: said holding walls include a first holding wall and a second holding wall; and said second holding wall is configured to define a cable passage channel for reaching of said at least one light module cable, which passes through said cable passage opening, to said longitudinal light module channel, and there is an intermediate space in a manner corresponding to said cable passage opening between said first holding wall and said second holding wall for reaching of said at least one light module cable to said cable passage channel through said longitudinal channel.

8. The domestic cooling device according to claim 2, wherein said external auxiliary carrier element has a bulged wall part which is adjacent to said cable passage opening and provided in a manner for guiding said at least one light module cable to said cable passage opening.

9. The domestic cooling device according to claim 5, wherein said external auxiliary carrier element has a casing part provided at a side which is opposite to a side where said connection extension which has said connection tab exist, and said casing part is embodied such that an electrical socket to which an end of said at least one light module cable is connected can be placed.

10. The domestic cooling device according to claim 9, wherein said casing part contains a socket casing wall provided such that said electrical socket, to which said end of said at least one light module cable is connected, can be rested.

11. The domestic cooling device according to claim 9, further comprising at least one stopper extension embodied at a channel input of said longitudinal channel which is adjacent to said casing part and shaped and dimensioned in a manner allowing passage of said at least one light module cable but in a manner preventing passage of said electrical socket.

12. The domestic cooling device according to claim 7, wherein said module carrier body has a longitudinal module carrier body channel which is adjacent to said light module channel and which extends generally parallel to said longitudinal light module channel.

13. The domestic cooling device according to claim 12, further comprising end caps disposed in said module carrier assembly, said end caps being retained in a removable manner to mutual end parts of said longitudinal module carrier body channel from one side and of said cable passage channel from another side and shaped and dimensioned in a manner providing at least partially covering said mutual end parts of said longitudinal module carrier body channel.

14. The domestic cooling device according to claim 13, wherein; each of said end caps contains a curved extension shaped and dimensioned in accordance with said cable passage channel, and in a condition where said end caps are retained to a related end part of said cable passage channel; and at a side of said curved extension facing said light module channel, there is a lateral opening formed therein and provided for extending said at least one light module cable to said light module channel.

15. The domestic cooling device according to claim 1, further comprising at least one adhesive element, wherein in order to fix said module carrier assembly to a lower planar surface of said planar base plate, said at least one adhesive element is disposed between said lower planar surface and a module carrier planar surface of said module carrier body facing said lower planar surface.